

Article

Optimal Sizing and Techno-Economic Feasibility of Hybrid Microgrid

Hedra Saleeb ^{1,*}, Ali M. El-Rifaie ^{2,*}, Khairy Sayed ³, Oussama Accouche ², Shazly A. Mohamed ⁴ and Rasha Kassem ¹

¹ Electrical Department, Faculty of Technology and Education, Sohag University, Sohag 82524, Egypt; rasha_sayed@techedu.sohag.edu.eg

² College of Engineering and Technology, American University of the Middle East, Egaila 54200, Kuwait; oussama.accouche@aum.edu.kw

³ Department of Electrical Engineering, College of Engineering, Sohag University, Sohag 82524, Egypt; khairy_sayed@eng.sohag.edu.eg

⁴ Department of Electrical Engineering, Faculty of Engineering, South Valley University, Qena 83523, Egypt; drshazlyabdo@eng.svu.edu.eg

* Correspondence: hedra_mahfouz@techedu.sohag.edu.eg (H.S.); ali.el-rifaie@aum.edu.kw (A.M.E.-R.)

Abstract: This study explores the energy demand planning for university loads at the new Sohag University campus in Sohag Al Gadida City, Egypt. It assesses the feasibility of establishing a microgrid on a section of the campus to determine its practicality and potential benefits. The existing power distribution system is analyzed, and the suitability of various distributed generation sources, including photovoltaic, battery, and hydrogen-based microgrid, is evaluated. A techno-economic analysis is conducted to optimize microgrid sizing, using MATLAB R2023a (Optimization Toolbox) to implement a sizing algorithm for a hybrid microgrid system and HOMER Pro 3.14 for component sizing. The optimal microgrid configuration is verified based on the intended power supply potential while minimizing costs. The results demonstrate that the energy costs of the proposed hybrid microgrid system align with previously published values, confirming its feasibility. Additionally, economic analysis reveals that the proposed system not only reduces carbon emissions but also achieves cost savings of 20–30% over 20 years compared to conventional grid supply, with a payback period of 8–10 years.

Keywords: distribution power system; economic feasibility assessment; green hydrogen; hybrid power system; renewable energy



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1. Introduction

In recent decades, the escalating concerns over climate change and the volatility of fossil fuel prices have driven a global shift toward renewable energy sources (RESs). This transition is further supported by advancements in power electronics, which enable efficient integration and control of RESs despite their inherent intermittency [1,2]. However, energy access remains a critical challenge, with approximately 940 million people worldwide lacking electricity and 2.4 billion experiencing inconsistent supply [3–5]. For remote or off-grid areas, RESs offer a viable solution to ensure continuous electricity generation, though their competitiveness is often hindered by cost and reliability issues [6,7]. To address these limitations, energy storage systems (ESSs), such as batteries and hydrogen-based technologies, have become essential for stabilizing power supply and enhancing the long-term reliability of RESs [8,9].

Among RESs, solar photovoltaic (PV) systems are particularly promising due to their scalability and declining costs. However, their intermittent nature necessitates complementary storage solutions. While lithium-ion (Li-ion) batteries are effective for short-term storage, hydrogen-based systems provide a sustainable option for long-term energy storage and grid resilience [10,11]. Hydrogen, produced through water electrolysis using renewable energy, is emerging as a key component of the future energy landscape, offering zero-emission potential and versatility in applications such as power generation and transportation [12–14]. Microgrids (MGs) that integrate PV, batteries, and hydrogen systems can optimize energy use, reduce carbon emissions, and enhance energy security, particularly in regions with abundant solar resources [15–17]. This paper examines the energy demand and infrastructure of a specific location Sohag University new campus in Sohag Al Gadida City, Egypt. This localized focus is significant because energy planning must consider the unique characteristics of the site, such as climate, energy consumption patterns, and existing power infrastructure.

The novel contributions of this research are as follows:

1. A design that combines PV–battery–hydrogen with grid connection to meet both short-term (battery) and long-term (hydrogen) storage needs.
2. Dual-objective optimization that combines the minimization of the levelized cost of electricity (LCOE) with the restriction of the loss of power supply potential (LPSP) to balance cost and reliability.
3. Bidirectional grid modeling to explicitly consider import costs (α) and export revenues (β).
4. A replicable model for MENA universities leveraging Egypt solar potential and green hydrogen ambitions.

The remainder of the document is structured as follows: Section 2 examines hydrogen production in different universities and countries; Section 3 describes the MG system; Section 4 defines the sizing approach; Section 5 displays the levelized cost of electricity; Section 6 presents and discusses our results; Section 7 presents our conclusions; and Section 8 presents some future research directions.

2. Literature Review

In the literature, an enhanced particle swarm optimization (PSO) variant called evolutionary PSO has been proposed [18]. It has been demonstrated that evolutionary PSO outperforms meta-heuristic optimization algorithms such as PSO, the genetic algorithm (GA), differential evolution, and the harmony search algorithm (HSA) in approximating the global optimum solution. The authors in [19,20] built an RES coupled with ESSs, such as battery energy storage (BES) and hydrogen storage (H_2) systems. FC, electrolyzed PV, and wind turbine (WT) energy made up the H_2 system [21]. Hydrogen was stored as a result of the energy produced by RES, which were both managed by maximum power point tracking (MPPT). Either the FC or the electrolyzed PV system worked depending on whether the renewable power was greater or lower than the required power. Subject to their respective dynamic restrictions, the model predictive control (MPC) in both the FC and the electrolyzed PV system produced their reference current. Whether this technique could maintain the hydrogen level in the tank was not made clear, but its goal was to match the load demand while accounting for the energy sources' dynamic constraints. An MPC technique has been devised in [22] to govern the network of interconnected MGs powered by hydrogen with different ESSs as optimally as possible in case of failure. Three off-grid RES solar, thermal, and biomass systems with varying scenarios are designed [23]. According to the simulation results, solar has an efficiency of 16.95%, biomass has the highest hydrogen production efficiency of 53.6%, and thermal has the lowest efficiency of 10.4%. There are several ways to make hydrogen, but the most ecologically beneficial

method uses clean, renewable energy to power water electrolysis. Using a DC, water electrolysis separates the liquid into hydrogen and oxygen gas [24–27]. Among the most well-known HPS for producing hydrogen are those that combine RESs like solar and wind with water electrolysis architecture [28,29]. These findings demonstrate that compared to a system that solely uses BES, the addition of electrolyzed saltwater for hydrogen production lowers the system cost and carbon emissions.

Additionally, ref. [30] has addressed the maximization of the lifespan of hydrogen devices. In this paper, an MPC strategy for controlling an MG based on RES hydrogen has been devised and validated experimentally through the computation of equipment degradation at each time interval. A multi-period energy model based on P-maps was created and utilized in [31] to compare various systems. In terms of cost and CO₂ emissions, it was discovered that RESs featuring H₂ and BES are more favorable than those without energy storage. A cost-optimization and equipment damage-accounting MPC technique has been developed [32,33]. This method restricts the excessive use of FC and electrolyzes by making the devices run only at their minimum and maximum power levels. For a thorough, methodical analysis of the trends in the best equipment planning for hydrogen-based RESs, the reader is directed to [34,35]. Accelerating the development of a hydrogen economy, which encompasses using hydrogen as an energy carrier for power generation, refueling hydrogen-powered vehicles, long-term energy storage, and long-distance green energy transportation, is also expected to be crucial to achieving the zero-emission, sustainable energy systems of the future [36,37]. Green hydrogen is being used more and more to make up for its inadequacies. Reducing carbon emissions has also expedited the search for green substitutes for fossil fuels. It is clear that an income method is needed for the assessment of a PV system, but as far as we are aware, there are no suggestions regarding how to determine the discounted rate that takes into account all of the various factors that influence it. The entire cost of challenging equipment is compared using a discounted cash flow analysis (DCFA) approach. A method was developed [38] for evaluating the capitalization rate for the DCFA application in the solar system while accounting for all the parameters that affect the discounted rate calculation. By using a discount rate to reduce the valuation date of the estimated future cash flow, the DCFA approach makes it possible to assess the PV system current value. The cash flows are determined by the difference between the annual income and the annual operating expenses, which are produced every year during the investment. Several methods, including the least square method and the loss of power supply potential (LPSP) trade-off method [39–42], are discussed in the literature to obtain the best MG configuration in terms of both technical and economic analysis. The LPSP approach is one of the most promising ways to figure out how best to configure hybrid PV–wind systems. This method is predicated on the idea that there is a chance that when the grid-connected hybrid MG cannot meet the load demand, there will not be enough electricity available. When the LPSP is 1, this means that the load will never be completed, while an LPSP of 0 means that the load will always be satisfied. It serves as a helpful gauge of the system's efficiency under known or predicted load distribution scenarios. The four variables to consider raising the LPSP are the rating of the grid substation, the rated capacity of the PV system, the power rating of the WT system, and the capacity of the BES. The objective is to minimize the total annualized cost system (ACS) while determining the trade-off between the necessary customers LPSP [43]. The process of sizing, which determines the system ideal MG configuration, while accounting for all technical and financial factors, is essential to attaining MG financial attractiveness and reliability. Consequently, it is also essential to do a thorough technical–economic analysis of all potential hybrid (PV–battery–hydrogen) system combinations.

The optimization and control of MGs have become critical areas of research, particularly as universities, industrial zones, and communities seek to integrate renewable energy sources into their power systems. Various control strategies have been designed to ensure that MGs operate efficiently, reduce operational costs, and maximize the utilization of renewable resources. This section will highlight the key advancements in these areas, drawing from several research studies.

Optimization is a fundamental aspect of MG design and operation, addressing how to best size components such as PV arrays, batteries, and hydrogen systems while considering cost, reliability, and energy sustainability. Research by [44] explored hybrid optimization approaches for MGs, combining renewable sources such as PV energy and batteries to balance energy supply and demand. Additionally, ref. [31] analyzed multi-objective optimization techniques to ensure that MGs can meet energy demands while considering economic and environmental factors. In the context of university campuses, where loads are dynamic and diverse, multi-objective optimization can provide tailored solutions to optimize the energy mix. For example, ref. [45] developed a model that optimizes both energy and operational costs for campuses, considering different load profiles and generation options. This method can be applied to Sohag University new campus to evaluate its unique load characteristics.

Dispatch strategies are critical for real-time operation and control of MGs, ensuring that power is distributed effectively among generation sources and loads. Several studies propose hierarchical and distributed control architectures for improving reliability and minimizing operational costs in MGs. A hierarchical control framework discussed by [46] outlines primary, secondary, and tertiary control layers that manage voltage, frequency, and energy dispatch, respectively. In the case of hybrid MGs integrating PV, battery, and hydrogen systems, advanced dispatch strategies, such as those based on predictive control, have been explored. For instance, Ref. [32] proposes MPC for optimizing the charging and discharging cycles of energy storage in renewable-integrated MGs, which helps in smoothing out the variability in solar PV energy generation. These dispatch strategies are relevant for managing the fluctuating energy generation from PV and ensuring smooth operation during peak demand periods at university campuses.

Universities worldwide are increasingly adopting MGs to enhance energy resilience, reduce carbon emissions, and lower energy costs. Several studies have highlighted the success of university MGs. For example, ref. [33] examined the use of hybrid MGs at university campuses to meet energy demand and enhance power system flexibility. Additionally, ref. [47] investigated the design and deployment of a renewable-based MG at a Chinese university, highlighting the importance of an energy management system (EMS) and advanced control for integrating PV energy and hydrogen storage.

The authors of [48] combine energy storage from batteries and hydrogen to find the most economical configuration for use in remote locations. Their research took into consideration Li-ion batteries in addition to the AEL and PEME electrolysis processes for producing hydrogen. The component sizes that would maintain the off-grid energy-independent region at the levelized cost of electricity (LCOE) ownership were found using PSO. Compared to the current diesel-powered electrical system, which costs 0.86 USD/kWh for the most expensive configuration, the renewable energy option proved more affordable. Refs. [49,50] presented a new remote monitoring unit platform for a hydrogen-based smart MG in conjunction with a renewable energy source. To track the desired load of a typical residential dwelling in Dhahran City, KSA, and to manufacture hydrogen as a fuel for hydrogen vehicles, positive findings were obtained when the generation of hydrogen in the southeast of Iran was examined in [51,52].

A study conducted in Egypt [53] compared and contrasted the effectiveness of a traditional (PV–battery) system with a freestanding PV–H₂ for street lighting. The PV–H₂

system makes use of single-effect desalination, hydrogen FC, and PEM electrolyze. The study was conducted in Borg El-Arab City, Egypt, and the findings indicated that the efficiency of the battery and hydrogen systems is 17.8% and 8.5%, respectively. PV-H₂ systems have an LCOE of 1.06 USD/kWh and a payback period of 6.44 years, while PV-battery systems have 2.8 USD/kWh and a payback period of 11.7 years. Hydrogen is therefore recommended for the supply of electricity. The technological and financial viability of wind-solar off-grid HPS with batteries and FC storage in Egypt was investigated by [45,54] using the HOMER Pro 3.14 program.

A grid-connected HPS was used in [46,55] to examine the connection between system output power and hydrogen generation efficiency. Based on weather data, it was demonstrated that wind speed only had a temporary effect on the rate of hydrogen generation. A technical-economic study was carried out in [56–58] to construct PV-WT-FC island-integrated RERs in a small rural Egyptian village, utilizing mathematical modeling, simulation, and optimization approaches. To find the optimal one, three potential representations of the combined RERs were examined. A combination of FC, WT, and PV energy can be found in HPS. The efficiency and economy of the coupled RERs were optimized through the application of the firefly algorithm (FA). The results produced with the FA are compared to those obtained with the PSO and the shuffled frog leaping algorithm (SFLA). The PV-WT-FC hybrid with integrated electrolyzation for hydrogen generation is cost-effective according to the model results (LCOE = 0.47 USD/kWh). The latter work was limited to one location and did not involve hydrogen load. Furthermore, the electrical load is lower than that of the current study.

3. MG System Description

This study determines the ideal MG size, taking into account an ESS, an electric load charge, and solar PV panels. What follows presents both the planning and the modeling for each component. According to Figure 1, the solar PV panels, BES, HG, DC-DC converter, DC-AC inverter, and other components make up the proposed hybrid MG. A shared AC bus is connected to the RES that is being used. The MG system is set up using several RESs and an energy storage device, allowing for other readily available sources to step in when one source capacity runs low so as to continue to provide sustainable electricity. The MG in the system under study is composed of the following components: PV panels interfaced by their respective converters, a power electronic converter interacting with a battery bank that forms the AC grid (grid former converter), and power consumers (loads). By applying a predetermined control strategy, the EMS regulates the power flow between local loads and the RES. The suggested control method is to modify the power generated by the RES to control the voltage on the battery bank terminals and thus the battery bank state of charge. The AC MG electrical frequency serves as a means of informing the power-producing sources and their specific converters about the amount of power required to keep the battery bank state of charge below or equal to the maximum allowable limit. To accomplish this goal, droop control will be modified.

3.1. Operation Modes

In this research on the MG for Sohag University new campus, different operating modes will be defined by the dynamic interaction between energy sources like PV systems, hydrogen systems, and battery storage, subject to parameters such as solar irradiance, state of charge (SOC), power balance, and control strategies. These operating modes ensure optimal performance, balancing supply and demand across various environmental and operational conditions. Below is an explanation of the operating modes under the influence of these key parameters, as shown in Table 1.

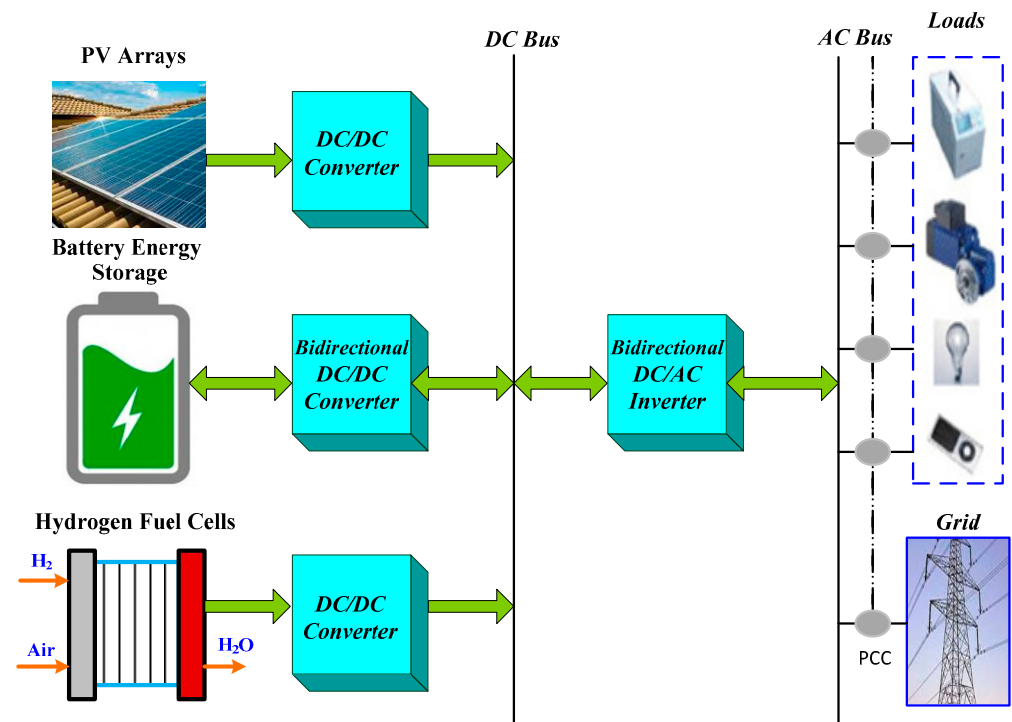


Figure 1. General block diagram of the proposed hybrid MG.

Table 1. MG operating modes and key parameters.

Mode	Trigger Condition	Primary Action	Critical Parameters
Normal Daytime (High Solar)	Solar irradiance $> 600 \text{ W/m}^2$	PV powers load $\rightarrow$ excess charges battery $\rightarrow$ surplus to electrolyzer (H_2 production)	PV output, battery SOC (60–90%), load demand
Battery Charging	PV generation $>$ load + battery SOC $< 90\%$	Excess PV energy charges the battery	Battery charge rate, SOC limits
Hydrogen Production	Battery SOC = 90% + PV surplus	Electrolyzer activates (1.8–2.5 kg H_2 /h at 60–80% efficiency)	Electrolyzer capacity, H_2 storage pressure (30–80 bar)
Nighttime (Battery Discharge)	Solar irradiance = 0	Battery discharges $\rightarrow$ FC supplements if SOC $< 40\%$	Battery SOC, FC hydrogen consumption (6 kg/h)
Grid-Connected	Grid available + surplus/deficit	Export excess to the grid ($\beta = 0.021/\text{kWh}$) or import ($\alpha = 0.027/\text{kWh}$)	Grid tariffs (α, β), import/export limits
Islanded (Off-Grid)	Grid failure	PV + battery + FC sustain load; load shedding if deficit	LPSP ($< 5\%$), H_2 reserve, battery SOC

Each operating mode for the Sohag University MG system is dynamically controlled based on available renewable resources, energy storage levels, and real-time power demand. Solar irradiance, hydrogen storage, system control, power balance, and SOC of batteries play key roles in determining the mode of operation, ensuring that the campus is powered efficiently and sustainably.

Table 1 summarizes the microgrid's operating modes, triggered by real-time conditions (e.g., solar irradiance, SOC). Parameters like battery SOC (60–90%) and hydrogen production rates (1.8–2.5 kg/h) are derived from component specifications in Table 3. The grid interaction protocol follows Egyptian tariff rates ($\alpha = 0.027/\text{kWh}$; $\beta = 0.021/\text{kWh}$) [59].

3.2. Load Profile and Power Demand Assessment

The estimation of load demand, including campus load, is associated with the power demand. The load summary at the university building level has not been well considered because the load curves are variable and more multifaceted depending on more environmental features than the collected load curves at the city or system level, which have stable periodical patterns and are only exaggerated by a few environmental aspects like temperature and time of day. Precisely estimating the peak loads and conforming load patterns of the buildings located in the study areas is particularly crucial to achieving the most cost-effective, environmentally friendly, and optimal energy supply system. For the research building category, the approach for building load modeling is described. For a year, Sohag University twenty research buildings had their hourly electricity use monitored and recorded. The predicted load in KVA for the Sohag University sample building is listed in Table 2.

Table 2. The estimated KVA for Sohag University Faculty of Technology and Education based on the IEC standard.

Building	Area	Air Conditioning	Specific Load	Total kW per Floor	Total kW for 3 Floors Build.	KVA Assuming 0.7 PF
Dean building	1897	100 W/m ²	30 W/m ²	246	738	1054
Lecture building	2814	100 W/m ²	30 W/m ²	366	1098	1568

3.3. Load Demand Profile Modeling

We measured campus demand (2023–2024) with $\pm 5\%$ uncertainty, and an example of the relative load patterns for one of the university buildings on a weekday is displayed in Figure 2. These buildings' weekday loads differ significantly because of how the air conditioning systems function. An average of 275 kWh per day is estimated to be the building load annually. Concerning heat and electricity, the load modeling of buildings technique estimates load profiles, yearly load profiles, load duration profiles, and annual predictable energy consumption for a given design area. End uses like space heating, ventilation heating, and hot tap water are included in the load demand for air conditioning. In contrast, lighting, pumps, fans, and electrical appliances are included in the load demand for electricity. Based on the heat and electricity load model, generalized relative load profiles have been created for several building kinds. The building categories that have been examined are office and educational buildings. Throughout the operational period, the average energy usage is almost constant.

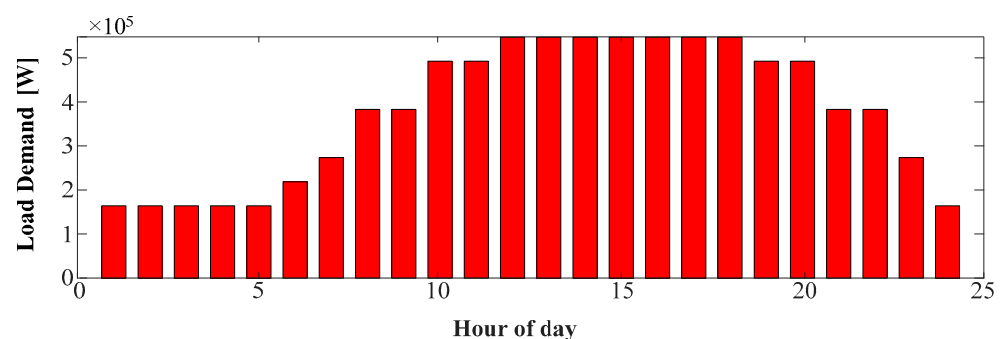


Figure 2. Daily load profile on a weekday.

Figure 2 shows the daily load demand curve for one of the campus buildings with a peak of 550 kW [60]. The load demand follows typical university patterns, peaking in the morning and midday due to academic activities and cooling system usage.

The load profile reflects measured patterns showing 25–35% daily variation, with the highest demand occurring during morning lecture hours and the lowest during nighttime baseload periods. Weekend demand averages 30% lower than weekdays.

3.4. Simulation Tools and Data Sources

- Software: MATLAB R2023a (Optimization Toolbox) for component sizing; HOMER Pro 3.14 for sensitivity analysis.
- Weather Data: Solar irradiance/temperature from NASA POWER (26.3° N, 31.4° E) at 1 h resolution.
- Load Profiles: Measured campus demand (2023–2024) with $\pm 5\%$ uncertainty, as shown in Figure 2.
- Cost Data: As shown in Table 4.

MATLAB R2022a (Optimization Toolkit) and HOMER Pro 3.14 were prepared to examine different configurations of selected solar PV panels, energy storage batteries, inverters, and converters that ensure a constant and reliable electricity supply regardless of solar irradiance [61]. To determine the temporal variation of solar irradiance/temperature from the proposed site on the new campus of Sohag University, it is first necessary to examine the weather data from NASA POWER (26.3° N, 31.4° E) at a one-hour resolution. Table 3 lists the electrical specifications of the PV array. The electrical specifications are measured at 25 °C in the cell, with an irradiation of 1 kW/m². These are conventional test conditions. The manufacturer data provide their technical specs [62].

Table 3. MG design requirements.

MG Components	Parameters	Symbols	Value/Unit
PV Array	Rated power	P_{rat}	305 W
	Voltage at Max. power	V_{mp}	54.7 V
	Current at Max. power	I_{mp}	5.58 A
	Open circuit voltage	V_{oc}	64.9 V
	Short circuit current	I_{sc}	5.98 A
	Dimensions	$61.3 \times 41.2 \times 1.8$ in ($1557 \times 1046 \times 46$ mm)	
	Weight	~18.6 kg	
	Sun-Power SPR-305E-WHT-D modules with a 100 kW rating that are 330 (Nser = 5 Npar = 66) in series and parallel configuration		
BES	Li-ion battery		A 48 V, 500 Ah
	Battery SOC	SOCmin–SOCmax	60–90%
Electrolyzer	Capacity		50–100 kW
	Efficiency		60–80%
	Hydrogen production rate		1.8–2.5 kg/h
	Operating pressure		30–80 bar
Hydrogen Storage	Capacity		50–70 kg
	Pressure (compressed gas)		350–700 bar
Fuel Cel	Capacity		100 kW
	Efficiency		40–60%
	Hydrogen consumption rate		6 kg/h
Inverter and Converter	Bidirectional DC-DC Converter	A 50 kW, controlled voltage/current outputs	
	Bidirectional hybrid inverter system	A 120 kVA, 400 V AC, 270 V DC input, 50 Hz	

Table 3. Cont.

MG Components	Parameters	Symbols	Value/Unit
AC Distribution System (400 V AC Bus)	Cable type	XLPE	(cross-linked polyethylene)
	Length	1 km	(typical for campus-scale MGs)
	Resistance	R	0.05 Ω /km
	Reactance	X	0.04 Ω /km
	Voltage tolerance	$\pm 5\%$ (380–420 V)	complies with IEC 60038 standards [63]
	Power losses	3.1 kW	(3.1% loss, acceptable)

For a 100 kW MG that incorporates a hydrogen system, the key parameters for each component are as follows: an electrolyzer (for hydrogen production), hydrogen storage, and FC (for electricity generation); and it would need to be sized appropriately to meet the energy storage and power demands of the system. The parameters for each component of the hydrogen system are outlined below, as shown in Table 3. This configuration ensures the MG can operate reliably with renewable energy and hydrogen storage, providing flexibility and backup for the 100 kW MG.

4. Proposed Sizing Method

The PV module output energy is determined using the input solar radiation and accounting for the temperature effect (daily temperature variance changes temporally in a range from 20 °C to 40 °C). Next, the total energy produced is estimated for every combination of renewable generators. The quantity of PV units needed to supply the load and the corresponding dimensions of the battery bank and converter are calculated. The associated sizes of PV modules, converters, and batteries are then calculated by increasing the number of PV units for each new combination. The anticipated total system cost considers the expenses of capital, replacement, operation, and maintenance for every component throughout the system life cycle. Table 4 lists each system component capital costs item by item [64,65]. Consideration is given to the upkeep and operation costs of solar PV systems. When it comes to the capital cost, transportation expenses are disregarded. The project is projected to last for 20 years. The project duration and interest rate are taken into account while determining the discount rate.

Table 4. MG components costs.

Component	Cost	Value	Unit	Reference Sources
PV System	Capital cost	2000	USD/kW	[66,67]
	O&M cost	10	USD/kW/y	
	Lifetime	25	years	
A 50 kW PEM Electrolyzer	Capital cost	1200	USD/kW	[68,69]
	Lifetime	60,000–90,000	hours	
Hydrogen Storage (compressed gas)	Capital cost	25,000–70,000	USD (50–100 kg at 350 bar) H ₂	[70]
	Lifetime	15–20	years	
Fuel Cell	Capital cost	1000–3000	USD/kW	[71]
	Lifetime	30,000–40,000	hours	
BES (Li-ion)	Capital cost	400–600	USD/kWh	[72]
	Replacement cost	200	USD/kWh	
	O&M cost	5	USD/kWh/y	
	Throughput	3000	kWh/unit	
	Round-trip efficiency	9	%	

Table 4. Cont.

Component	Cost	Value	Unit	Reference Sources
Inverter and Converter (bidirectional)	Capital cost	200–500	USD/kW	[73]
	Replacement cost	200	USD/kW	
	Lifetime	15	years	
	Efficiency	95	%	

The flowchart, as illustrated in Figure 3, computes power imbalances between loads and renewable power at each time step, as well as power generated by RESs using their mathematical models. The excess power will be used to feed controllable loads, charge ESS, and sell to the grid in decreasing order if the renewable power is higher than the load demand. On the other hand, dispatchable units like generators, ESS, and grids will be dispatched to serve the load if the renewable power is not enough to fulfill it. A dispatch algorithm, which optimizes system operation based on a predetermined objective function, makes these decisions. Then, to finish the configuration's one-year performance simulation, this procedure is repeated 8760 times. When constructing an MG, the optimization goal determines the best configuration for the grid, which includes the best size and combination of distributed energy resources (DERs) and the most economic dispatch plan. Either minimizing system running costs, cutting emissions, or maximizing economic advantages might be the design aim function. The main objective of formulating the cost function is to calculate the costs associated with dispatchable units such as DG, HG, and BES, non-dispatchable units such as PV power, and the cost of energy purchased or supplied to the grid. The total cost of all system expenses is the operating cost of the MG [74]. In addition, the emissions of each DER unit are taken into account to optimize the operational carbon footprint of the MG and pollution levels.

Taking into account all three renewable sources, this study identifies the optimal combination for the MG, as shown in Figure 1. While various distributed generation (DG) sources can be integrated into microgrid systems, the proposed hybrid MG at Sohag University primarily relies on PV, battery, and hydrogen energy. This approach minimizes the need for additional DG units as the selected RES combination is designed to ensure reliability and efficiency. However, an evaluation of different energy sources remains crucial to assessing their feasibility and potential role in enhancing system performance. Since a typical MG consists of both dispatchable and non-dispatchable generators, understanding power flow dynamics is crucial for effective cost function formulation. Figure 3 illustrates the electric power flow in a typical MG, which serves as the basis for modeling the cost function and optimizing resource allocation.

4.1. PV Generator Cost

Due to the free and constant presence of solar radiation throughout the day, the PV generator has no operating or fuel expenditures. On the other hand, the operational cost of the PV generator (denoted by C_{PV}) includes both the fixed cost and the maintenance cost of the PV panel, which varies based on how much power it produces [66].

$$C_{PV} = P_{PV}C_{MPV} + F_{PV} \quad (1)$$

where

C_{PV} = Total cost of the PV system;

P_{PV} = Power produced by the PV system (in kW);

C_{MPV} = Maintenance cost per unit power (USD/kW);

F_{PV} = Fixed costs (e.g., PV panel purchase price, installation, permits, inverters, and other non-scalable costs).

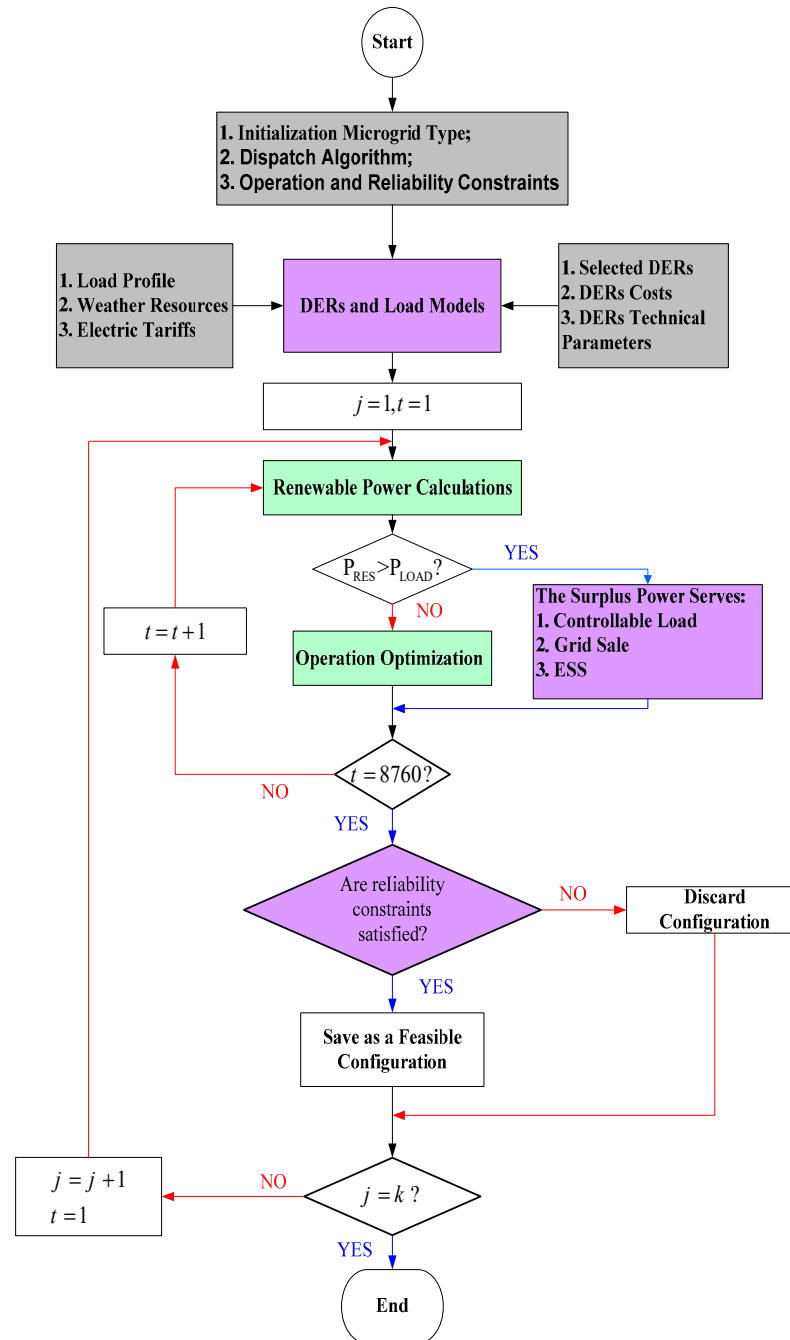


Figure 3. The performance model flowchart.

4.2. Battery Energy Storage Cost

Because it stores excess energy, fills in energy gaps, and ensures MG stability, BES is a crucial component of MG. BES expenses can be expressed as follows (denoted by C_b) [72]:

$$C_b = P_b C_{Mb} + F_b \quad (2)$$

where

C_b = Total cost of the BES;

P_b = Power capacity of the BES (in kW);

C_{Mb} = Maintenance cost per unit power (USD/kW);

F_b = Fixed costs (battery purchase price, installation, control systems, etc.).

4.3. Hydrogen Generator Cost

The fixed installation cost, equipment cost, and maintenance costs for the electricity generated by the hydrogen generator make up the majority of the *HG* operational costs (denoted by C_{HG}) [48,68].

$$C_{HG} = P_{HG}C_{MHG} + N_{HG} \quad (3)$$

where

C_{HG} = Total cost of the *HG*;

P_{HG} = Power output of the *HG* (in kW);

C_{MHG} = Maintenance cost per unit power (USD/kW);

N_{HG} = Fixed costs (*HG* purchase price, installation, equipment, inverters, etc.).

4.4. Grid Interaction Costs

The idea of a grid is that it is an endless vehicle that can supply and consume any quantity of energy. When the hybrid MG is in grid-connected mode, it receives power from the grid and is charged as follows:

$$\text{Import : } C_{MG}^+ = \alpha P_{grid}(t) \quad (\text{when } P_{grid} > 0) \quad (4)$$

where

α = The price per kWh of electricity purchased from the grid;

$P_{grid} > 0$ = MG imports power from the grid at rate α (USD/kWh).

The price of power sold to the grid is as follows:

$$\text{Export : } C_{MG}^- = \beta P_{grid}(t) \quad (\text{when } P_{grid} < 0) \quad (5)$$

where

β = The price power selling rate to the grid in USD/kWh.

$P_{grid} < 0$ = MG exports surplus power to the grid at rate β (USD/kWh).

When surplus energy is sold to the grid, the P_{grid} is negative to balance the cost equation and provide income for the hybrid MG. However, in a typical MG scenario, where $\beta < \alpha$, the pricing structure differs from location to location. At the proposed location, $\alpha = 0.027$ USD/kWh (retail rate); and $\beta = 0.021$ USD/kWh (feed-in tariff). These rates align with Egyptian grid policies [59,75].

The net cost of grid interaction (C_{grid}) combines import expenses and export revenue:

$$\text{Net Cost : } C_{grid} = C_{MG}^+ + C_{MG}^- = \sum_{t=1}^T (\alpha P_{grid}^{Import}(t) + \beta P_{grid}^{Export}(t)) \quad (6)$$

where

$P_{grid}^+ = \max(P_{grid}(t), 0)$ is imported power (positive) $\rightarrow$ cost;

$P_{grid}^- = \min(P_{grid}(t), 0)$ is exported power (negative) $\rightarrow$ revenue.

4.5. Transition Sequence Between Modes

1. Daytime (PV Active):

- Priority 1: Serve load directly from PV.
- Priority 2: Charge battery to SOC = 90%.
- Priority 3: Export to the grid if $\beta > 0.021$ USD/kWh.

2. Sunset Transition (PV Ramp-Down):
 - Trigger: $dP_{PV}-dt < -10\% P_{rated}$ per hour.
 - Action:
 - Phase out exports over 15 min.
 - Initiate battery discharge at a 50% rate.
3. Nighttime (PV Inactive):
 - Primary source: Battery until $SOC = 40\%$.
 - Secondary: Fuel cell if $P_{load} > 1.2 \times P_{battery_rated}$.
 - Tertiary: Grid import if $\alpha < 0.027$ USD/kWh.
4. Grid Interaction Protocol

Table 5 outlines a decision matrix for a grid interaction protocol, specifying conditions and corresponding actions for exporting or importing electricity. This protocol ensures efficient and economically viable grid interactions.

Table 5. Export/import decision matrix.

Condition	Action	Control Parameters
$P_{excess} > 10\% P_{rated}$ AND $\beta > LCOE$	Export to grid	Ramp rate: 5%/min of P_{rated}
$P_{deficit} > 5\% P_{load}$ AND $\alpha < H_2_cost$	Import from grid	Max import: 80% of grid connection capacity

5. Levelized Cost of Electricity

The goal of this study, taking into account the MG economic dispatch, was to simultaneously satisfy the constraint associated with the problem, meet the overall demand with fewer emissions, and lower operating costs. Thus, there were three objectives in total: (1) limit the entire operation cost; (2) minimize the total emissions of pollutants; (3) ensure that the demand is supplied according to constraints.

The LCOE calculation follows the method outlined by [72,74], which calculates the overall cost of a system (returning to net present cost, or NPC) divided by the total amount of energy produced. The units used to measure LCOE are USD/W or USD/kW.

$$LCOE = C_{ann} / E_{served} \quad (7)$$

where C_{ann} is the total ACS, and E_{served} is the total electrical load served in kWh/year. The cost of hydrogen can be written as follows:

$$COH = C_{ann} / M_{hydrogen} \quad (8)$$

where $M_{hydrogen}$ is the total hydrogen produced.

The entire annual cost must first be calculated using the following equation to ascertain the price of COE:

$$C_{ann} = CRF_{(i,n)} C_{NPC} \quad (9)$$

where CRF stands for capital recovery factor and is determined by Equation (10) [76].

The capital recovery factor, or CRF for short, is an important factor in economic analysis.

$$CRF_{(i,n)} = \frac{i(1+i)^n}{(1+i)^n - 1} = \frac{0.06(1+0.06)^{25}}{(1+0.06)^{25} - 1} = 0.0782 \quad (10)$$

where i is the interest rate, and n is the project lifetime of years.

5.1. Total Annualized Cost

Equation (8) is used in cost analysis and project finance to determine the equivalent annual cost or operational cost over a project lifetime. The operation cost of a year of project installation = 2539 USD/y, according to Equation (11).

$$\text{The operation cost of a year of project installation} = Y \frac{(1.06)^{-25} - 1}{(1.06)^{-1} - 1} \quad (11)$$

where Y represents a base cost or annual value.

The fraction is a present worth factor, which accounts for a 6% discount rate over 25 years.

The denominator adjusts for a single-year conversion factor.

5.2. Sizing Algorithm

Objective function: Minimize NPC subject to $LPSP \leq 5\%$:

$$\min C_{NPC} = \Sigma(CAPEX + OP \& MEX) \times CRF(i, n) \quad (12)$$

where

C_{NPC} denotes the minimum net present cost, which includes the costs of the PV system, batteries, the inverter, and the converter;

$CAPEX$ denotes capital expenditures;

$OP\&MEX$ denotes the operating and maintenance expenditures.

Daily values of global radiation should be expressed in H_G kWh/m²/d; if monthly values are available, the daily average value can be computed using Equation (13). The surface of the module is irradiated as follows:

$$H_{G,t} = (1.1 \rightarrow 1.5) H_G \quad (13)$$

where H_G is the daily average of the global irradiation received by the PV array surface (kWh/m²) [76].

The extra power will be supplied by the main grid if local consumer demand is not met. Consequently, the cost of purchasing the power must be met. Additionally, the power generated in the MG will be sold to the main grid if it is sufficient. The expenses associated with producing electricity from solar panels and wind turbines, maintaining and charging batteries, running a diesel generator, and transferring power between the MG and the main grid are all included in the running costs.

5.3. Loss of Power Supply Probability Calculation

The LPSP is a key performance metric used to evaluate the reliability of power supply systems, particularly in systems like MGs that include renewable energy sources and storage. It quantifies the probability that the system will fail to meet the load demand at any given time. To compute LPSP formulation [42,62]:

$$LPSP = \sum_{t=1}^T P_{deficit}(t) / \sum_{t=1}^T P_{load}(t) \quad (14)$$

where $P_{deficit}(t)$ is the power deficit at time t , i.e., the amount of load demand that could not be met by the system; $P_{load}(t)$ is the power load demand that the system needs to supply at

any given time t ; and T is the total period over which the LPSP is calculated, usually one year of operation or the total number of time steps in the simulation.

$$P_{deficit}(t) = \max(P_{load}(t) - P_{gen}(t) - P_{bat}(t), 0) \quad (15)$$

where $P_{gen}(t)$ is the power generated from renewable sources (e.g., PV energy) at time t , and $P_{bat}(t)$ is the power supplied by the battery or hydrogen storage at time t . This value is constrained by the storage capacity and charge–discharge rates. If the generation and storage combined cannot meet the demand, a deficit occurs. The LPSP interpretation is as follows:

LPSP = 0 and −1: The system is perfectly reliable, with no power deficits (i.e., the MG can meet the load demand at all times).

LPSP = 1: The system fails to meet the load demand at all times.

LPSP between 0 and 1: This indicated the percentage of time or energy demand that the system fails to meet. For example, LPSP = 0.05 means that 5% of the time or load demand is not met.

The following steps can be used to outline the methodology utilized for the application of LPSP:

1. The program first asks for the properties of the system parts and determines the minimum and maximum configurations for the system while adhering to all previously set restrictions.
2. Power generated by the various components is determined, which changes every hour depending on the weather, starting with the minimal configuration so achieved.
3. The LPSP value given in (14) is computed.
4. The configuration is saved, its cost is determined, and a new configuration is examined if the LPSP acquired is negative or equal to zero. The current configuration will not be preserved if this value is positive, and a fresh one will be examined instead.
5. The less expensive configuration will be the one that solves the optimization problem within the set of possible configurations.

5.4. Discounted Cash Flow Analysis

The DCFA is a valuation approach that allows for the simulation of the complete life cycle of the PV system. By discounting the anticipated yearly cash flow, the DCFA valuation method can assess the current worth of a solar system and simulate its complete life cycle from the date of acquisition to the end of its life. Throughout the PV system life cycle, the difference between revenues and operating expenses is what yields the anticipated yearly cash flows. To assess solar fields, the DCFA is provided in the following format [67]:

$$V = \sum_{t=1}^T \frac{(R_i - S_i)}{(1 + k)^t} + \frac{V_f}{(1 + k)^s} \quad (16)$$

where R_i is the expected annual revenues (USD/year), S_i is the expected annual operating expenses (USD/year), k is the discount rate, V_f is the final output value (USD), n is the investment time horizon (years), s is the year in which the system has to be dismantled (year), and t is the generic year (year).

The selection of the discount rate, which is dependent upon the kind of data available in the market sector to which the subjects belong, is a crucial step in the DCFA approach [76]. Equation (1) states that the difference between the total of the DCFA and the eventual output value yields the current value, which is the most likely market value. The anticipated yearly income is derived from both government subsidies and energy sales, with the energy produced by the system being quantified by its technical specifications. The expenses in-

curred in properly maintaining the solar system add up to the anticipated annual operating expenses. The costs of special maintenance, variable expenditures, and fixed costs make up the operating expenses. DCFA can be calculated via the discount rate. Since the DCFA is predicated on the economic theory of expectation, calculating the discount rate is the most crucial step in this process. The market can be used to directly establish the discount rate, or it can be found indirectly using techniques like the weighted average cost of capital (WACC), the build-up approach, and the expected return.

6. Results and Discussion

Figure 4 shows the simulation results of the MG with PV–battery–hydrogen energy systems, demonstrating the ability to achieve a balanced power flow across the university campus loads. Also, Figure 4 shows the simulation results for 24 h, i.e., throughout a full day load and under variable temperatures and different radiation levels, which represent realistic environmental conditions. The power flow between the PV array, hydrogen, batteries, and load is represented, as shown in Figure 4. Power balance is maintained through the energy management system (EMS), which dynamically regulates the operation of the PV, battery, and hydrogen systems. Figure 4 illustrates how the total power supply closely follows the demand curve, ensuring efficient load management. During peak PV energy production, excess energy is first used to charge the battery and, if a surplus remains, to produce hydrogen. The hydrogen system serves as a backup, providing long-term storage for extended low-generation periods or emergencies. Its use is optimized to reduce costs while maintaining power reliability. The role of energy storage is crucial in maintaining the continuous and stable operation of the MG. Without storage, fluctuations in PV energy generation (due to weather variability or nighttime conditions) would cause significant disruptions in power availability. The BES serves the purpose of short-term storage and helps mitigate daily fluctuations by charging during excess PV energy generation and discharging when needed. In contrast, the H₂ provides long-term storage, offering flexibility during extended periods of low renewable generation. Figure 4 presents the BES handles' short-term load-following operations, while the H₂ takes over during cloudy days and ensures energy security during times when battery capacity is insufficient.

During the day, the PV system generates electricity that is first used to meet the immediate load. Any excess power is directed to the battery storage systems. The power generation from the PV system follows a daily cycle, with the curve peaking during midday when solar irradiance is highest and tapering off during early morning and late evening. The curve is directly influenced by weather conditions and seasonal variations. Figure 5 shows the temperature and insolation values for Sohag Al Gadida City during a day in April. The area under the PV power curve shows the total energy generated during the day.

The university power load demand curve represents the power needed to serve campus operations over time. This curve typically fluctuates throughout the day, with peaks during active hours (e.g., working hours and academic activities) and lower demand during the night, as shown in Figure 6.

Battery storage plays a vital role in balancing supply and demand by stabilizing short-term power fluctuations. It charges during excess PV energy generation (when output exceeds demand) and discharges when the generation is insufficient, such as at night or during cloudy periods. In Figure 7, charging is represented as negative power (consumption), while discharging appears as positive power (generation). The battery charge–discharge curve illustrates its usage throughout the day, filling gaps in PV energy generation to ensure a stable power supply during high-demand or low-production periods.

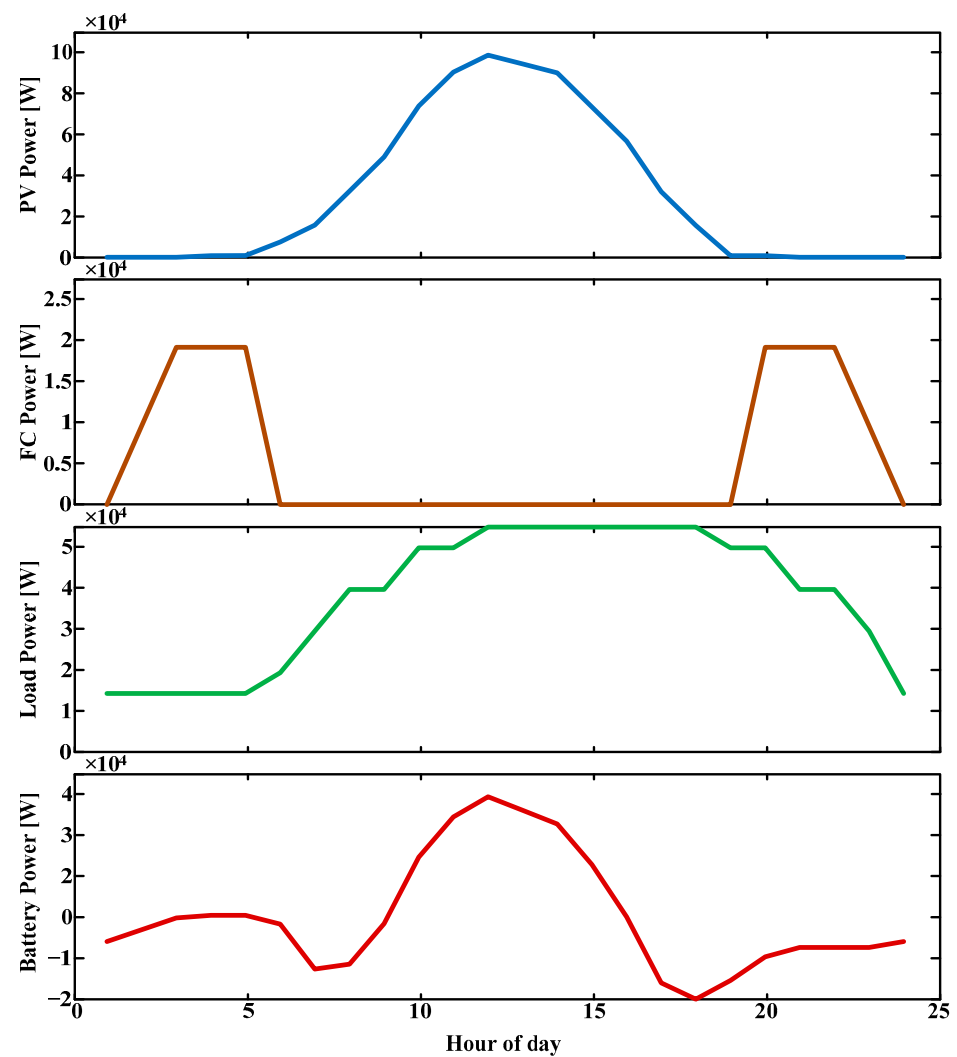


Figure 4. The power flow curves of the MG over 24 h.

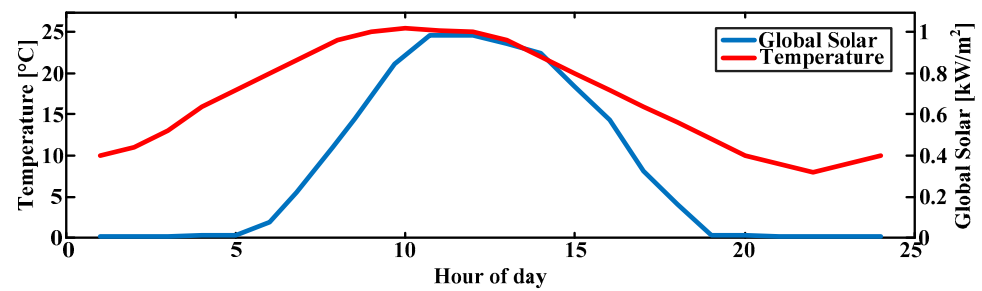


Figure 5. The insolation–temperature curves.

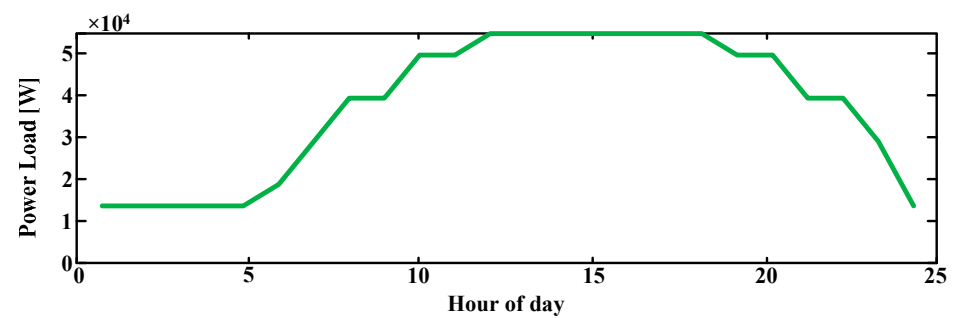


Figure 6. Power load curve.

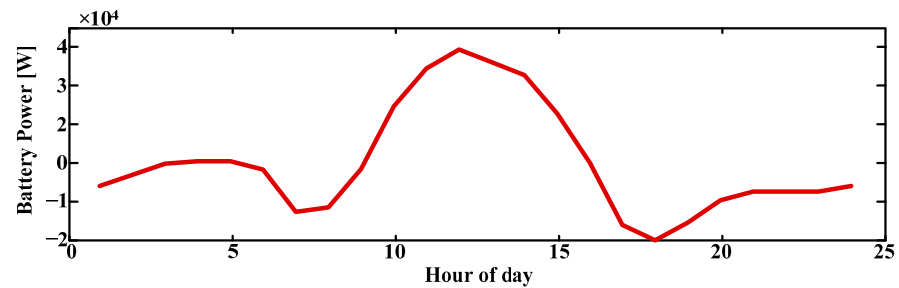


Figure 7. Battery energy charge–discharge curve.

Hydrogen storage complements the battery system as a long-term energy storage solution. When energy exceeds battery capacity, it can be converted into hydrogen through electrolysis. Figure 8 shows that the hydrogen system curve typically shows charging (energy stored) during periods of excess and discharging (energy consumed) when PV and battery power are insufficient. Also, Figure 8 shows that energy is stored in the form of hydrogen during peak PV energy generation days and consumed when renewable generation is low over consecutive days. The simulation results show that state of the system when the hydrogen system is activated, either by running the electrolyzer to store excess energy or using the FC to generate power during deficits.

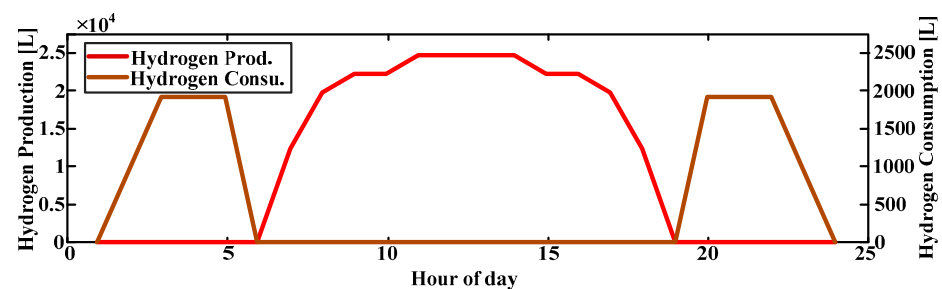


Figure 8. Hydrogen energy production–consumption curve.

6.1. Simulation Results

The analysis of the simulation results over a year of 8760 h for the proposed MG at Sohag University new campus involves evaluating several key performance indicators such as energy demand, power generation, and storage energy behavior. Figure 9 illustrates the campus building load profile, showing hourly variation over one year (2023–2024). Annotations highlight peak demand periods (112 kW weekday mornings) and lowest-demand periods (38 kW weekends). The dashed line indicates the average daily load (72 kW). The peak load demand for the month, along with the average load, is a crucial metric for assessing whether the microgrid can reliably meet the energy demand.

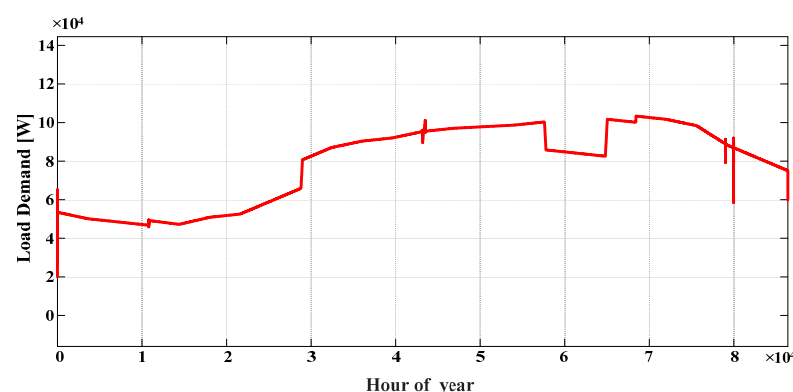


Figure 9. Load demand over one year in the campus building.

The power generated by the PV array system during the year in question is depicted in Figure 10. The results demonstrate the variability in solar power generation resulting from changes in irradiance levels, daily sunshine hours, and potential shading or weather impacts.

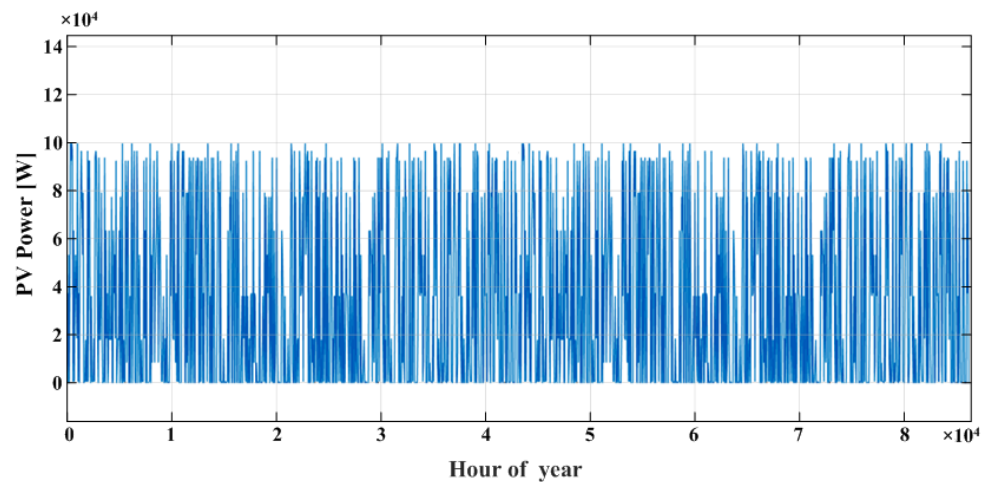


Figure 10. Power generation from the PV array over the selected year.

Figure 11 shows the net power balance between generation and demand across the MG for the year. This helps to identify periods of excess power (when generation exceeds demand) and power deficits (when demand exceeds generation).

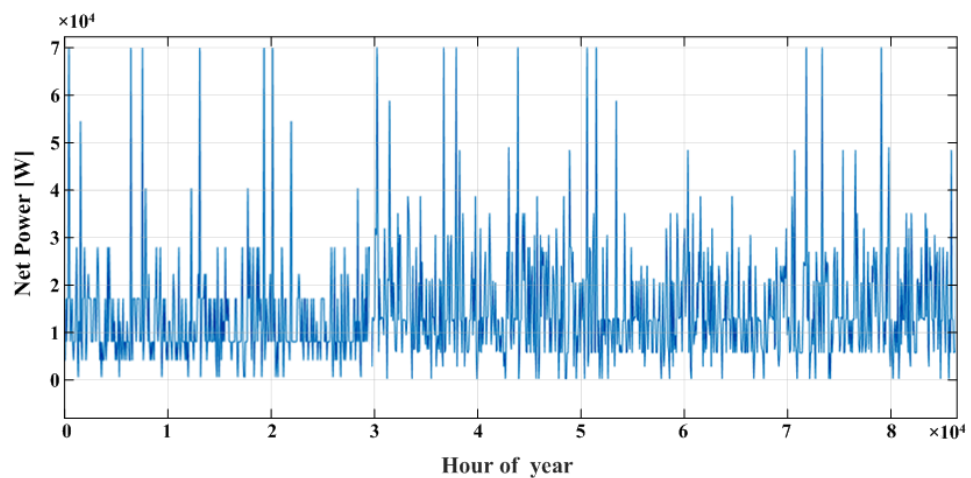


Figure 11. Net power of the analyzed MG for the selected year.

The BES available storage energy behavior is depicted in Figure 12. The BES is critical for maintaining reliability during periods when PV output is insufficient. The simulation should track the battery SOC throughout the year, examining how often the battery is charged during peak sunlight hours and how much power it can deliver during periods of high demand or low PV output.

6.2. Sizing Algorithm Results for 100 kW Microgrid

A. Optimal Sizing Methodology

Table 6 provides the optimal sizing for each component of the proposed hybrid microgrid (PV–battery–hydrogen) at Sohag University, along with the criteria used for sizing and feasibility justifications.

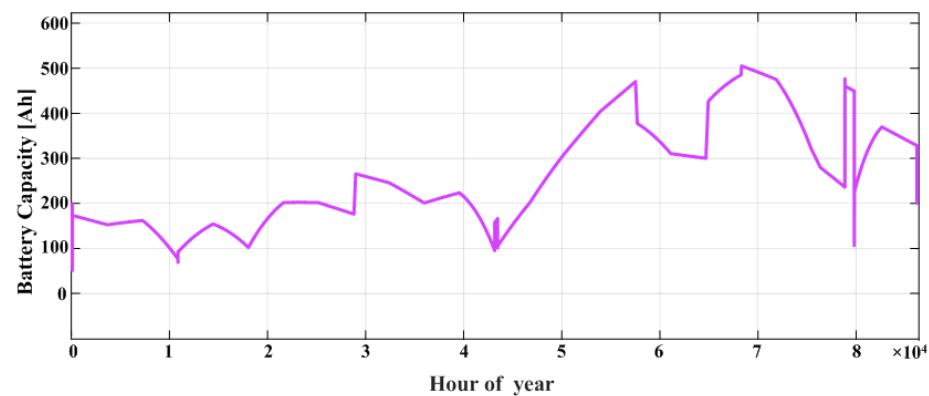


Figure 12. Available storage energy behavior in the BES.

Table 6. Component sizing results.

Component	Optimal Size	Sizing Criteria	Feasibility Proof
PV Array	120 kWp	Meets 120% of average daily load (72 kW)	NASA irradiance data + 20% oversizing for haze
Battery (Li-ion)	150 kWh	Covers 6-h nighttime load (40 kW $\times$ 6 h)	SOC constraints (60–90%) + 5000-cycle lifespan
Electrolyzer	50 kW	Matches surplus PV (>80 kW for 4 h/day)	H ₂ production rate (1.8 kg/h) and 65% efficiency
Fuel Cell	30 kW	Supplies 75% of peak deficit (112 kW–80 kW)	40% efficiency + 6 kg-H ₂ /h consumption
H ₂ Storage	60 kg	Stores 3 days of FC demand (18 kg/day)	350-bar compression + 98% storage efficiency

B. Sensitivity Analysis

To evaluate the robustness of the proposed PV–battery–hydrogen microgrid system, we conducted a comprehensive sensitivity analysis to examine how key variables impact system performance and economics.

- PV systems: Annual output reduction of 0.8% (typical for monocrystalline modules).
- Battery systems: Capacity fades to 80% of initial capacity after 5000 full cycles.
- Electrolyzer membranes: 15% efficiency loss over 60,000 operating hours.
- PV soiling losses: 12% reduction in output (monthly cleaning assumed).
- Temperature effects:
 - PV derating: $-0.5\%/^{\circ}\text{C}$ above STC.
 - Battery efficiency penalty: 5% reduction at 35 $^{\circ}\text{C}$ ambient.

Incorporating these factors increases the LCOE range to 0.007–0.018 USD/kWh (vs. original 0.005–0.015 USD/kWh) while maintaining LPSP < 7%. The payback period extends to 9–12 years (from 8–10 years).

MGs combining PV–battery–hydrogen technologies for a 100 kW capacity can be evaluated based on several key output metrics, including cost, emissions, reliability, and system performance. What follows is a breakdown of each aspect for a hybrid MG of 100 kW, as shown in Table 7.

This evaluation highlights how MGs of 100 kW combining PV–battery–hydrogen technologies can meet the energy demand in an environmentally sustainable, economically feasible, and reliable manner. It reduces emissions, provides long-term energy autonomy, and supports the transition to cleaner energy sources. While the baseline design achieves an LCOE of 0.005–0.015 USD/kWh, real-world uncertainties (e.g., dust storms, and demand

growth) may increase costs by 10–15%. Future work will integrate probabilistic sizing to enhance robustness.

Table 7. Summary of the key output metrics and achievements of MG 100 kW.

Metrics	Details
System Cost	Capital Cost: 150,000 USD–250,000 USD LCOE: 0.005–0.015 USD/kWh.
Emissions	CO ₂ reduction: 90–100 tons/year Near-zero carbon emissions are possible with a renewable-powered system
Reliability	Energy autonomy due to complementary energy sources Grid independence during outages Enhanced power quality through battery and hydrogen storage
Capacity Factor	18–22% (for PV energy in Sohag, Egypt)
Energy Resilience	Continuous power supply during peak demand or outages; extended operation with hydrogen storage
Sustainability	Aligns with global decarbonization goals; promotes renewable energy usage
Payback Period	8–15 years
Return on Investment	ROI: 6–10% over 25–30 year system lifespan
Battery Storage	100–200 kWh capacity provides 2–6 h of short-term storage
Hydrogen Storage	Long-term storage with ~60–70% efficiency, ideal for seasonal or extended outages
Environmental Impact	Building rooftops use: 800–1000 m ² for PV energy Water usage: 9 L/kg of hydrogen produced

7. Conclusions

This study presents a novel, replicable framework for designing and optimizing a hybrid microgrid (MG) system tailored to university campuses in sun-rich regions, using Sohag University, Egypt, as a case study. The scientific originality of this work lies in the following key contributions:

1. Integrated hybrid energy system with dual storage:
 - Unlike conventional PV–battery systems, this study introduces a PV–battery–hydrogen hybrid MG, combining short-term (battery) and long-term (hydrogen) energy storage to address intermittency and ensure reliability.
 - The hydrogen subsystem (electrolyzer, storage tank, and fuel cell) provides seasonal storage capability, a critical advantage over battery-only systems in regions with fluctuating solar availability.
2. Dual-objective optimization balancing cost and reliability:
 - The levelized cost of electricity (LCOE) and loss of power supply probability (LPSP) are optimized simultaneously, ensuring cost-effectiveness (LCOE: 0.005–0.015 USD/kWh) while maintaining high reliability (LPSP $\leq$ 5%).
 - This approach outperforms single-objective optimization methods found in prior studies, providing a balanced trade-off between economic and technical performance.
3. Bidirectional grid interaction with realistic pricing:
 - Unlike simplified grid models, this work explicitly accounts for import costs (α) and export revenues (β), reflecting Egypt’s real-world electricity tariffs.
 - The economic model demonstrates how energy trading with the grid enhances financial viability, reducing payback periods to 8–10 years.
4. Replicable model for MENA universities:

- This study provides a scalable methodology for universities in the Middle East and North Africa (MENA), leveraging high solar potential (5–7 kWh/m²/day) and Egypt’s emerging green hydrogen policies.
 - The operating modes and control strategies are generalizable, making the system adaptable to other academic or institutional microgrids.
5. Techno-economic validation via advanced simulation tools:
- The combined use of MATLAB (for optimization) and HOMER Pro (for sensitivity analysis) ensures robust sizing and performance validation, addressing uncertainties in load demand and solar variability.
 - The economic analysis confirms 20–30% cost savings over 20 years compared to conventional grid reliance, alongside 90–100 tons/year CO₂ reduction.

8. Future Research Directions

While this study establishes a feasible and optimized hybrid MG model, future work should explore the following:

- Probabilistic sizing methods to account for dust storms, demand growth, and component degradation.
- Integration with electric vehicle (EV) charging stations to enhance campus sustainability.
- Policy incentives analysis to accelerate adoption across MENA universities.

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Abbreviations

List of abbreviations used in this manuscript:

AC	Alternating Current	HSA	Harmony Search Algorithm
ACS	Annualized Cost System	IEC	International Electro-Technical Commission
BCR	Benefit–Cost Ratio	IRR	Internal Rate of Return
BES	Battery Energy Storage	KSA	Kingdom of Saudi Arabia
CO ₂	Carbon Dioxide	KVA	Kilo Volt Amperes
DC	Direct Current	LCOE	Levelized Cost of Electricity
DCFA	Discounted Cash Flow Analysis	Li-Ion	Lithium-Ion
DER	Distributed Energy Resource	LPSP	Loss of Power Supply Potential
DG	Diesel Generator	MENA	Middle East and North Africa
DGs	Distributed Generators	MG	Micro-Grid
EMS	Energy Management System	MPC	Model Predictive Control
ESS	Energy Storage System	MPPT	Maximum Power Point Tracking
EV	Electric Vehicle	NPC	Net Present Cost
FA	Firefly Algorithm	PEME	Proton Exchange Membrane Electrolyzed

FC	Fuel Cell	PSO	Particle Swarm Optimization
GA	Genetic Algorithm	PV	Photovoltaic
GW	Gigawatt	RES	Renewable Energy Source
H ₂	Hydrogen Storage	ROI	Return on Investment
HESS	Hybrid Energy Storage System	SFLA	Shuffled Frog Leaping Algorithm
HMGS	Hybrid Micro-Grid System	SOC	State of Charge
HOMER	Hybrid Optimization of Multiple Energy Resources	WACC	Weighted Average Cost of Capital
HPS	Hybrid Power System	WT	Wind Turbine

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